

LAB REPORT

Search for mutations responsible for antibiotic resistance in *E. coli*.

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ARTICLE HISTORY

Compiled October 13, 2024

ABSTRACT

During this project, we made quality control of the raw reads from Illumina sequencing, improved the quality of raw data by removing low-quality parts, and mapped our reads to reference sequences by finding optimal local alignment in the reference. Our next task was to find out mutations in our reads and whether they change any proteins in *E. coli* DNA. For this, we made an SNP annotation and visualized the results. 5 SNP mutations were found. In analyzing these mutations, 3 of them were found to be responsible for giving antibiotic resistance in *E. coli*.

KEYWORDS

E. coli; antibiotic resistance

1. Introduction

Escherichia coli (*E. coli*) is one of the most common Gram-negative bacteria. *E. coli* is usually present in the human and animal body as a component of the microflora and has an important role in digestion and absorption. On the contrary, it can be a disease-causing agent in almost any part of the human body, for example causing gastrointestinal infections, diarrhea, infections of the urinary tract, and many others. Because of this, *E. coli* resistance to antibiotics is a major worldwide clinical problem.

Infections caused by *E. coli* are successfully treated with β -lactam antibiotics that contain a β -lactam ring in the chemical structure of the antibiotic. These include penicillin derivatives, such as penicillins, cephalosporins, and cephamycins, monobactams, carbapenems, and carbacephem. The majority of β -lactam antibiotics bind to and inactivate the essential PBPs (penicillin-binding proteins) responsible for the formation of mature peptidoglycan in the bacterial cell wall. PBPs are responsible for the last stages of the assembly of bacterial cell walls. These enzymes are targets for beta-lactam antibiotics. Two PBP activities include dd-transpeptidase and DD-carboxypeptidase activities that cross-link the cell wall and trim the peptidoglycan, which is the main component of the cell wall, to amino acids.[1]

Currently, however, the usefulness of β -lactams is seriously compromised because of the large number of hydrolytic enzymes - β -lactamases or ESBL - extended-spectrum β -lactamases expressed by bacteria and because of the loss of outer membrane porins, which reduces drug entry. The threat is exacerbated by the extremely flexible genome of *E. coli* and the propensity to spread resistance through horizontal gene transfer and

clonal spread. [2]

2. Methods

The work was performed using a Linux Bash Shell. During the project, we work with the [reference sequence of E. coli from NCBI FTP in .fna and .gff formats](#) and [Illumina sequencing data of an E. coli resistant to ampicillin in .fastq format](#). Firstly, we analyzed the statistics of Illumina sequencing reads from sequencing of E.coli with the FastQC tool (QC - quality control)[3]. Results of quality control show that we have low per-base sequence quality, which means that some bases have very low coverage. Consequently, we filtered bases with the Trimmomatic[4], using the following parameters: LEADING:20, TRAILING:20, SLIDINGWINDOW:10:20, MINLEN:20. Repeated check with FastQC tool showed that we now have a satisfactory quality of the sequence. In the next step, we aligned the reference sequence and sequence of ampicillin-resistant E.coli with the BWA-MEM (Burrow-Wheeler Aligner), which is optimized for long next-generation sequencing reads and makes use of the Burrow-Wheeler Transform. Then, we compress and sort alignment data with the SAMtools. Finally, we automatically annotate SNP (Single Nucleus Polymorphism) with SnpEff[5] (short “SNP effect”). We visualized the resulting annotation using the IGV browser[6]. As a result, we got 5 mutations which were analyzed further.

3. Results:

As a result of the trimming, some reads were dropped. The results are in Table 1.

Table 1. Table 1. A number of lines and readings.

	Before	After
Lines	1 823 504	1 785 036
Reading	455 876	446 259

We mapped the reads to the reference and scanned to detect positions containing mutations. Five SNPs were detected: 3 missense variants, 1 upstream gene variant and 1 synonymous variant in chromosome NC_000913.3. They are presented in Table 2.

Table 2. Table 2. Description of found SNPs

gene	mutation	substitution	position	coding	product
ftsI	missense	Ala544Gly	93043	protein	peptidoglycan DD-transpeptidase
acrB	missense	Gln569Leu	482698	protein	multidrug efflux pump RND permease
rybA	upstream gene variant		852762	protein	small RNA
envZ	missense	Val241Gly	3535147	protein	sensor histidine kinase
rsgA	synonymous variant	Ala252Ala	4390754	protein	ribosome small subunit-dependent GTPase A

4. Discussion:

We identified 3 mutations leading to amino acid substitution. This mutation in

the *Escherichia coli* FtsI gene (also called PBP3) is a transpeptidase necessary for synthesis of peptidoglycan in the septal division, and it is one of several proteins that localize in the septal ring. Amino acid substitutions in this gene can reduce the localization of the septal ring, a multi-protein complex that plays an important role in cell division, or does not support cell division at all. In addition, the mutation can cause a change in the orientation of FtsI to the peptidoglycan wall, so that transpeptidation becomes impossible, but because the active site is not affected, penicillin-binding is not impaired. [7]

A second mutation in *acrB* gene. As a rule, Gram-negative bacteria have a much higher level of resistance to various antibiotics than Gram-positive bacteria. One reason is how effective the outer membrane as a barrier is. Gram-negative bacteria require the additional contribution of pumps for the outflow of many drugs to reach their characteristic resistance levels. Resistance Nodulation and cell Division (RND) family pumps are particularly effective in forming resistance because they form a triple complex with the periplasmic proteins belonging to the membrane fusion protein (MFP) family and the outer membrane channels. [8] One of the most studied members of the RND is the internal membrane drug antiporter AcrB, part of the main outflow system AcrAB(Z)-TolC. AcrB is structurally a homotrimer, and each of its protomers consists of a transmembrane domain, a porter domain, and a funnel domain. We can assume that the amino acid substitution results in a change in the structure of one of the domains, thus causing outflow-mediated antibiotic resistance. For example, one of the lactam antibiotics, carbapenems, usually binds to a substrate recognition site close to the funnel domain. [9]

Also, mutation in *envZ* gene may be the reason for antibiotic resistance and that is why. EnvZ, an *E. coli* histidine kinase, responds to changes in osmolarity in the medium and regulates the phosphorylation state of the transcription factor OmpR, which controls the expression levels of the outer membrane porin proteins OmpF and OmpC. [10] Because there was no frameshift or insertion of stop codons in our data, we can assume that *envZ* mutations may be responsible for the decrease in ompCF expression, which, as in the previous case, affects membrane permeability.[2]

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